

POTHOLE PRIORITY PREDICTOR

An AI Driven Road Maintenance Decision Support System

Institute: International Institute of Medical Science & Technology Council (IIMSTC)

Cohort: DSML Cohort 12

Guide(s) : Suhas Gajanana

Student(s) : 1. Bharath M 2. Dhanyashree K 3. Gagan HR 4. Gurugula Surya Chenna Siva Sankar Reddy 5. Madan BD 6. Nandish R 7. Peeyush Suryan 8. Rajendra Gowda M 9. Rohan ME 10. Saniya Kausar T 11. Shreelatha G 12. Sowjanya BM 13. Sukhi R 14. Tasbeeha F

Keywords: *Pothole Detection, YOLOv11, Image Segmentation, Pavement Distress Index, RICE Priority Model, Pavement Condition Index, Flask, Road Maintenance, Karnataka PWD, IRC:SP:72.*

Introduction :

Potholes remain a critical road safety crisis in India, causing 2,385 deaths and Rs. 10,000 crore in annual repair costs in 2024 alone. Over the 2020–2024 period, 9,438 deaths and 23,056 accidents have been directly attributed to pothole-related incidents - a 53% increase in fatalities over five years. Existing systems rely entirely on manual inspection and reactive maintenance, with no capability to analyse formation factors, predict deterioration, or prioritise repairs intelligently.

The Pothole Priority Predictor addresses this gap through a unified AI-driven decision support pipeline. By leveraging YOLOv11 object detection and segmentation, the system automatically detects potholes from road images, estimates their severity using the Pavement Distress Index (PDI), ranks them by repair priority using a RICE-based scoring model, and estimates repair costs using the Karnataka PWD Schedule of Rates 2023 and IRC:SP:72 standards. The system is deployed as a Flask REST API with an interactive frontend dashboard and a Bangalore zone heatmap powered by Leaflet.js.

No existing commercial or civic system provides an integrated pipeline that tells road authorities which pothole to fix first, backed by AI-driven severity scoring, deterioration forecasting, and cost estimation. This project directly addresses that gap, designed specifically for Indian municipal road authorities such as BBMP (Bruhat Bengaluru Mahanagara Palike).

Literature survey:

Sl.No.	Author(s) / Source	Methodology	Key Contribution	Gap / Our Addition
1	iWatchRoadv2, 2025	YOLOv8 + GPS mapping	Automated detection with geospatial mapping	We add severity scoring, deterioration prediction, priority ranking, and cost estimation
2	Enhanced YOLOv8 + Drone, 2026	YOLOv8 + attention + FPN, drone imagery	Improved detection in low light and bad weather	We use YOLOv11s segmentation for pixel-level area and IRC-based repair cost estimation
3	YOLOv8 + MiDaS, 2025	YOLOv8 + MiDaS depth estimation	Automates dimension measurement and repair budgeting	We use Street View imagery and live traffic APIs for dynamic priority ranking
4	PoDAS Sensor Network, 2025	Ultrasonic sensors + accelerometer + GPS	Low-cost real-time detection (~80% accuracy)	We use computer vision — no custom hardware, deployable on any camera
5	ML + Pavement Condition, 2025	ML on pavement condition indicators	Predicts pothole occurrence locations	We add RICE priority ranking and PCI deterioration tracking per zone
6	YOLOv11-s + Traffic Data, 2025	YOLOv11s + traffic volume API	Priority order based on traffic volume	We extend with segmentation masks, PDI severity, IRC:SP:72 cost, and ASTM D6433 deterioration

Objectives :

1. Detect potholes from road images using YOLOv11 object detection and segmentation models trained on a 9,727-image Roboflow dataset.
2. Assess severity using the Pavement Distress Index (PDI) combining pixel area density, shape complexity (perimeter²/area), and model confidence with camera-angle correction.
3. Prioritise repairs using a RICE-based scoring model (Reach × Impact × Confidence / Effort) adapted for road infrastructure, producing a 0–100 priority index.
4. Predict deterioration using Pavement Condition Index (PCI) tracking from ASTM D6433 across multiple scans of the same zone to support proactive maintenance.
5. Suggest repair types and estimate costs using Karnataka PWD Schedule of Rates 2023 and IRC:SP:72 standards (pothole patching at Rs.850/sqm to full reconstruction at Rs.9,500/sqm).

6. Provide an interactive Bangalore zone heatmap (Leaflet.js) for geographic visualisation of pothole density across BBMP zones.
7. Deploy a Flask REST API backend with a real-time detection dashboard for municipal road authority use.

Methodology :

The Pothole Priority Predictor follows a structured end-to-end AI pipeline from data collection through model inference and cost estimation to deployment.

Dataset and training:

The dataset was sourced from Roboflow Universe, comprising 9,727 images pre-annotated with bounding boxes and segmentation masks in YOLOv11 format (train: 6,809 | val: 1,944 | test: 974). Roboflow handled all preprocessing — augmentation (flip, brightness, blur), resizing to 640×640, and data.yaml generation. The primary detection model was trained using YOLOv11n on Google Colab (T4 GPU) for 50 epochs, batch size 16, confidence threshold 0.25, and early stopping patience 10.

Severity Assessment – Pavement Distress Index(PDI)

Severity scoring uses PDI from Arya et al. (2021) with Indian road weights from Mandal et al. (2023): $PDI = (Area\ Density \times 0.45) + (Confidence \times 0.35) + (Shape\ Complexity \times 0.20)$. Area density is computed from pixel-level segmentation masks; shape complexity captures irregular geometries via $perimeter^2/area$. A dashcam camera-angle correction factor of 1.35 accounts for the ~26% area compression in downward-facing imagery.

Priority Scoring and Cost Estimation:

Priority scoring uses a RICE model adapted for road infrastructure, producing a 0–100 priority index. Repair cost estimation follows IRC:SP:72 and Karnataka PWD SOR 2023 — ranging from surface patching (Rs.850/sqm) to full depth reclamation (Rs.9,500/sqm). Deterioration tracking uses ASTM D6433 PCI, classifying zones from Good (≥ 85) to Failed (< 25) to enable proactive repair scheduling before catastrophic failure.

System Architecture:

The Flask REST API (/predict) accepts image uploads, runs YOLOv11 inference, computes PDI severity, RICE priority score, repair type, and repair cost, and returns GPS coordinates for heatmap integration. The frontend dashboard provides four modules: Detection, Analytics, Heatmap (Leaflet.js, Bangalore zone grid), and Scan History.

Result and conclusion :

The YOLOv11n detection model was evaluated on the 974-image held-out test set. The trained model achieved strong detection performance as shown below:

Metric	Score
mAP@50	88.3%
mAP@50-95	65.8%
Precision	86.1%
Recall	81.0%

Following initial bounding-box training, fine-tuning of the segmentation model (lr=0.001, 20 additional epochs) enabled pixel-level mask prediction for accurate area-based PDI computation. The hybrid detection-segmentation pipeline, combined with the RICE priority engine and IRC:SP:72 cost estimation module, delivers an end-to-end decision support system for municipal road maintenance.

In conclusion, the Pothole Priority Predictor successfully automates road condition assessment through intelligent detection, severity scoring, and repair prioritisation. The system demonstrates how open-source AI models (YOLOv11, PyTorch) and lightweight web frameworks (Flask, Leaflet.js) can be combined to deliver production-ready civic infrastructure tools — with the potential to reduce manual road inspection costs significantly and enable proactive maintenance scheduling for urban road authorities like BBMP.

References :

- [1] iWatchRoadv2: Pothole Detection, Geospatial Mapping, and Intelligent Road Governance, 2025.
- [2] A Deep Learning Framework for Real-Time Pothole Detection from Combined Drone Imagery Using Enhanced YOLOv8, 2026.
- [3] Pothole Detection and Cost Estimation using YOLOv8 + MiDaS Depth Estimation, 2025.
- [4] Pothole Detection and Analysis System (PoDAS) for Real-Time Data Using Sensor Networks, 2025.
- [5] Pothole Prediction Based on Machine Learning and Pavement Condition Indicators, 2025.
- [6] YOLOv11-s-Based Pothole Detection Model with Integrated Traffic Data for Maintenance Decision Support, 2025.
- [7] D. Arya et al., RDD2020: An Annotated Image Dataset for Automatic Road Damage Detection Using Deep Learning, Data in Brief, 2021.
- [8] V. Mandal et al., Deep Learning Based Road Damage Detection for Indian Roads, IEEE Access, 2023.
- [9] ASTM D6433, Standard Practice for Roads and Parking Lots Pavement Condition Index Surveys.
- [10] IRC:SP:72, Guidelines for the Design of Flexible Pavements for Low Volume Rural Roads, Indian Roads Congress, 2015.
- [11] Karnataka PWD Schedule of Rates 2023, Public Works Department, Government of Karnataka.
- [12] A. Wang et al., YOLOv11: Real-Time Object Detection with Enhanced Architecture, Ultralytics, 2024.

Future scope :

The future scope of this project includes:

1. Mobile Application — A native Android/iOS app enabling field engineers to upload pothole images on-site and receive instant priority scores and repair cost estimates.
2. Real-Time Video Stream Processing — Extend the pipeline to process dashcam video feeds frame-by-frame for continuous road condition monitoring during municipal patrol vehicles.
3. City-Wide Cloud Deployment (AWS/Azure) — Scale the Flask backend to a cloud infrastructure capable of handling simultaneous uploads from multiple BBMP ward engineers.
4. Automated MoRTH Reporting — Generate monthly road condition reports in Ministry of Road Transport and Highways (MoRTH) compliance format directly from the system dashboard.

5. Multi-City Generalisation — Extend the PWD rate tables and zone coordinates to other Indian cities (Chennai, Pune, Hyderabad) with city-specific IRC and PWD schedule integration.